

API DATA RETRIEVAL & ANALYSIS

Complete User Manual

A Plain-Language Guide for All Users

Powered by [OpenAQ v3 API](#) · [openair R Package](#) · [R Shiny](#)

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Chapter 1: Introduction

1.1 What Is API Data Retrieval & Analysis?

API Data Retrieval & Analysis is a web application that connects you to the world's largest open-source air quality database (OpenAQ). Without writing a single line of code, you can:

- Find air quality monitoring stations anywhere on Earth
- Download real-time and historical pollution measurements
- Create professional charts and visualisations
- Export your data to Excel, CSV, or PDF for further use

The application is built using the R programming language and runs in your web browser. You do not need to install any other software apart from R and RStudio and an internet connection and a free OpenAQ account.

1.2 Who Is This Manual For?

This manual is written for anyone who wants to explore air quality data, regardless of their technical background. Whether you are a student, researcher, journalist, public health professional, or simply a curious citizen, this guide will walk you through every feature step by step.

NOTE: No programming knowledge is required. Every action in this application is performed through buttons, menus, and drop-down lists.

1.3 What Is OpenAQ?

OpenAQ (Open Air Quality) is a non-profit organisation that aggregates air quality data from government monitoring networks, research institutions, and low-cost sensor deployments worldwide. As of 2026, the platform provides access to data from over 100 countries, covering pollutants including:

Pollutant	Symbol	Unit	Health Relevance
Fine Particulate Matter	PM2.5	µg/m ³	Respiratory & cardiovascular disease
Coarse Particulate Matter	PM10	µg/m ³	Lung irritation
Nitrogen Dioxide	NO2	µg/m ³	Traffic emissions, lung damage
Ozone	O3	µg/m ³	Photochemical smog, asthma
Sulphur Dioxide	SO2	µg/m ³	Industrial emissions, acid rain
Carbon Monoxide	CO	µg/m ³	Combustion, reduces blood oxygen
Black Carbon	BC	µg/m ³	Diesel soot, climate forcing
Temperature	Temp	°C	Meteorological context
Relative Humidity	RH	%	Meteorological context

All data on the OpenAQ platform is freely available under the Creative Commons CC BY 4.0 licence, meaning you can use, share, and adapt it freely as long as you credit the original source.

1.4 System Requirements

To use the API Data Retrieval & Analysis tool, you need:

- A modern web browser (Chrome, Firefox, Edge, or Safari updated within the last two years)
- A stable internet connection
- A free OpenAQ account (to obtain an API key)

- No software installation required when using the hosted version if you have R and RStudio

TIP: For the best experience, use Google Chrome or Microsoft Edge. Some older browsers may display the dark terminal theme incorrectly.

Chapter 2: Getting Started

2.1 Creating Your OpenAQ Account

You need a free API key to retrieve data from OpenAQ. Follow these steps:

1. Open your web browser and go to: <https://explore.openaq.org>
2. Click the Sign-Up button in the top-right corner.
3. Enter your email address and choose a password.
4. Confirm your email address by clicking the link in the verification email.
5. Once logged in, click your profile icon (top-right) and select API Keys.
6. Click Generate New Key. Give it a name such as 'My API Data Retrieval & Analysis Key'.
7. Copy the key; it is a long string of letters and numbers. Store it somewhere safe, such as a text file on your computer.

IMPORTANT: Your API key is like a password. Do not share it publicly. If it is compromised, you can generate a new one from the same settings page. Exceeding API rate limits repeatedly can lead to a temporary or permanent ban from the OpenAQ API. [→ Read more here](#). To maintain uninterrupted access, plan your API requests carefully

2.2 The Application Layout

When you open API Data Retrieval & Analysis, you will see the main interface divided into three areas:

Area	Location	Purpose
Left Sidebar	Vertical panel on the left	Navigation menu and API key connection
Main Content	Large central area	The active tab's content maps, data and charts
Header Bar	Top strip	App title, theme toggle, and links

The sidebar contains six navigation tabs:

- **Location Explorer:** find monitoring stations on a map
- **Fetch Data:** download measurements from selected sensors
- **Analysis:** create charts and visualisations
- **Export:** download your data and plots
- **Help & Guide:** quick reference and troubleshooting

- **About:** application information and acknowledgements

2.3 Connecting Your API Key

Before you can retrieve any data, you must connect your API key. This takes about ten seconds.

Step-by-step:

8. Locate the API Key field at the top of the left sidebar. It shows a password box labelled 'OpenAQ API Key'.
9. Click inside the box and paste your API key (Ctrl+V on Windows, Cmd+V on Mac).
10. Click the green Connect button.
11. Wait two to three seconds. A status badge will appear at the bottom of the sidebar.
12. If the badge shows a green 'Connected' message, you are ready to use the app.
13. If you see a red 'Not connected' message, double-check your key and try again.

TIP: If the connection fails, ensure you copied the entire key with no spaces at the beginning or end. The key is typically 40-60 characters long.

2.4 The Dark / Light Theme Toggle

API Data Retrieval & Analysis uses a dark terminal theme by default (black background, cyan and green text). If you prefer a brighter display, you can switch to a light theme at any time.

- Click the small crescent moon button (☾ Dark) in the top-right corner of the header.
- The theme switches immediately to a light background.
- To switch back, click the same button again (now showing ☀ Light).
- Your preference is saved automatically and remembered the next time you open the app.

NOTE: Switching themes does not affect your data or any charts you have already generated.

Chapter 3: Location Explorer

The Location Explorer is your starting point. It lets you find air quality monitoring stations anywhere in the world using an interactive map and four different search methods.

3.1 Understanding the Layout

The Location Explorer tab has three rows of panels:

- Top row - Search panel (left), Map Options panel (centre), Quick Start panel (right)
- Middle row - Interactive map (left, large) and Found Locations list (right)
- Bottom row - Locations Table (collapsed by default)

3.2 Search Method 1: Search by City or Name

This is the easiest way to find stations if you know the name of a city or a monitoring station.

How to use it:

14. Click the City / Name tab inside the Search panel.
15. Type a city name (e.g. 'Accra', 'Delhi', 'London') or a station name into the text box.
16. Optionally set the Max Results number (default is 30; increase if you want more results).
17. Optionally filter by parameter using the drop-down (e.g. show only stations measuring PM2.5).
18. Click the Search button.
19. Results appear as blue dots on the map and as cards in the Found Locations list.

TIP: If your city name returns no results, try the country name, a nearby larger city, or switch to the Coordinates search method.

3.3 Search Method 2: Search by Coordinates

Use this method when you know the exact latitude and longitude of your area of interest, or when you want to find stations near a point you click on the map.

How to use it:

20. Click the Coordinates tab in the Search panel.

21. Enter a latitude and longitude, OR simply click anywhere on the map - the coordinates update automatically.
22. Adjust the Radius slider (1 to 25 km). This sets how wide an area to search around your point.
23. Click Search Nearby.

IMPORTANT: The OpenAQ API limits the search radius to 25 km. If you need a wider area, use the Bounding Box method instead.

3.4 Search Method 3: Bounding Box

A bounding box lets you define a rectangular region by drawing it on the map or by entering four coordinates manually. This is useful for searching a whole city district, province, or country.

Using the draw tool:

24. Click the Coordinates tab in the Search panel, then switch to the Bounding Box tab.
25. On the map, click the rectangle draw icon in the top-left toolbar.
26. Click and drag to draw a rectangle over your area of interest.
27. Release the mouse. The four coordinate boxes in the panel fill in automatically.
28. Click Search Box.

Entering coordinates manually:

29. Type the Min Lon (westernmost longitude), Max Lon (easternmost), Min Lat (southernmost latitude), and Max Lat (northernmost) directly into the four boxes.
30. Click Search Box.

3.5 Search Method 4: Search by Country

This method retrieves all stations in an entire country. It is best used when you want a broad overview.

How to use it:

31. Click the Country tab in the Search panel.
32. Start typing a country name in the drop-down box. Results filter as you type.
33. Select the country and click Search Country.

NOTE: Large countries like India or China may return hundreds of stations. Use the Locations Table at the bottom of the page to sort and filter them.

3.6 The Interactive Map

The map is the central feature of the Location Explorer. Here is what you can do with it:

Action	What It Does
Click a blue dot	Selects that station; loads its sensors; opens a popup with the station name and ID
Click anywhere else on the map	Sets the coordinates in the Coordinates search tab
Scroll wheel/pinch	Zooms in and out
Click and drag	Pans the map
Draw rectangle	Sets the bounding box for the Bounding Box search
Globe button (bottom left)	Resets the map to the world view

You can change the map background style using the Map Style drop-down in the Map Options panel. Available styles include CartoDB Light, CartoDB Dark, OpenStreetMap, Satellite, and Topographic.

Tick the Cluster markers checkbox to group nearby stations into numbered clusters when zoomed out. This is helpful when many stations are close together.

3.7 The Found Locations List

After a successful search, the right panel shows up to 30 location cards. Each card displays:

- The station name (bold, top of card)
- The city or district and country
- The location ID (a number used internally)
- The date of the most recent data update
- A small badge showing the number of sensors at that station

Click any card to select that station. The selected card turns green and the map highlights it with a green ring. The app automatically loads the sensors for that station and takes you to the Fetch Data tab.

3.8 The Locations Table

Below the map, there is a collapsed panel labelled Locations Table. Click the arrow to expand it. This table shows all found stations in a sortable, searchable grid - useful when you have many results and want to filter by country, sensor count, or last update date.

TIP: Click any column header to sort by that column. Use the Search box above the table to filter by any text value.

Chapter 4: Fetch Data

The Fetch Data tab is where you download measurements from the sensors at your chosen station. It is divided into three sections across the top row: the selected location summary, the sensor selector, and the fetch control.

4.1 Selected Location Panel

The left panel confirms which station is currently selected. It shows:

- Station name
- City/district
- Country
- Location ID
- Latitude and longitude
- Latest readings from each sensor (if available)

If this panel says 'No location selected', go back to the Location Explorer tab and click a station.

4.2 Sensors Panel - Choosing What to Download

Once a station is selected, the middle panel shows all the sensors at that station as a list of checkboxes. Each checkbox label shows:

- The parameter name (e.g. pm25, no2, temperature)
- The measurement unit in brackets (e.g. [$\mu\text{g}/\text{m}^3$])
- The sensor ID number
- The most recent recorded value

Selecting sensors:

- All sensors are pre-ticked when the list first loads.
- Click individual checkboxes to select or deselect specific sensors.
- Use the All button to tick every sensor quickly.
- Use the None button to untick all sensors.

TIP: For most analyses, start by fetching all sensors. You can always filter or focus on specific parameters during the Analysis step.

4.3 Fetch Controls

The right panel contains three settings that control what data you download:

Temporal Aggregation

This controls the time resolution of the data:

Option	Meaning	Best For
Raw measurements	Every individual reading as recorded	Short time periods; sensor diagnostics
Hourly averages	One value per hour per sensor	Most standard analyses; recommended default
Daily averages	One value per day per sensor	Long time periods; trend analysis

Date Range

Click the calendar icon to set a start and end date. The maximum available period depends on when the station began reporting.

NOTE: Requesting raw measurements over a long time period (more than a few weeks) may return very large datasets and take longer to download.

Max Records Per Sensor

This sets an upper limit on the number of rows to retrieve per sensor. The default is 1,000 rows. Increase this number if you need a longer time period or if your data appears to be cut off.

4.4 Fetching the Data

34. Check that all three controls are configured.
35. Click the large green Fetch Data button.
36. A spinning status indicator appears. Wait while the app contacts the OpenAQ API.
37. When complete, a green status message shows the number of rows and parameters retrieved.
38. The Data Preview table below updates automatically.

TIP: If you see a yellow warning saying 'No measurements returned', try widening the date range or selecting a different aggregation level. Some stations only report data infrequently.

4.5 The Data Preview Table

The lower half of the Fetch Data tab shows the downloaded data as a table. The columns are:

- date - the timestamp of each measurement (UTC)

- One column for each selected sensor parameter (e.g. pm25, NO2, temperature)

This is the openair-ready wide format - each row is one point in time, and each column is one pollutant or variable. This format is required by the Analysis tools.

Reading the completeness bars:

Above the table, coloured progress bars show what percentage of time slots have valid data for each parameter:

- Green bar - more than 80% of values are present (excellent)
- Yellow bar - 50-80% of values are present (acceptable)
- Red bar - fewer than 50% of values are present (patchy; consider widening the date range)

4.6 Downloading the Data

Below the preview table, four download buttons let you save the data to your computer:

Button	What It Saves
CSV	A plain-text comma-separated file. Opens in Excel, Google Sheets, or any text editor.
RDS	An R data file. Use this if you want to reload the data directly into R.
Excel	A Microsoft Excel .xlsx file. Preserve column formatting.
ZIP	A compressed archive containing both the CSV and RDS files together.

Chapter 5: Analysis

The Analysis tab is the heart of the application. It gives you access to 15 professional-grade air quality charts powered by the openair R package, which is the industry standard for atmospheric data analysis.

5.1 Preparing Your Data

Before generating any chart, you must prepare your data. The Data Source panel at the top of the tab lets you choose between two sources:

Option A - Use Fetched OpenAQ Data (recommended)

39. Select 'Use fetched OpenAQ data' (this is selected by default).
40. Check the status box - it should say how many rows and parameters are ready.
41. Click the Prepare Data button.
42. A confirmation message appears. You are now ready to create charts.

Option B - Upload Your Own CSV File

43. Select 'Upload CSV file'.
44. Click the file selector and choose a CSV file from your computer.
45. A column mapping panel appears. Select which column contains the date/time values.
46. Optionally map wind speed (ws) and wind direction (wd) columns if your data has them.
47. Choose the date format that matches your file from the drop-down.
48. Click Prepare Data.

IMPORTANT: Your CSV must have at least one date/time column and at least one numeric pollutant column. Column names can be anything - you will map them in the interface.

5.2 The Plot Type Selector

The Plot Controls panel on the left shows a grouped drop-down menu with 15 plot types organised into five categories. Select a plot type to see its description and a set of controls tailored to that chart.

5.3 Summary & Time Category

Summary Plot

The Summary Plot provides a complete overview of your dataset in a single view. The top panel shows time-series lines for each pollutant averaged to your chosen period (hourly, daily, monthly, etc.). The bottom panel shows data completeness bars so you can immediately see if any parameters have gaps.

Controls: Averaging period (min, hour, day, week, month, year) and an optional period-OK indicator.

TIP: Start with the Summary Plot every time you fetch new data. It gives you an instant quality check before any deeper analysis.

Time Plot

The Time Plot draws one or more time-series lines on the same chart. It is ideal for comparing two pollutants side by side over time.

Controls: Select which pollutants to show, averaging period, whether the Y-axis scale is shared or free per variable, facet (conditioning) type, and a smooth-line toggle.

Time Variation

This produces a four-panel chart showing how concentrations vary by: hour of day, day of week, month of year, and a combined hour-by-day-of-week matrix. It is excellent for identifying rush-hour peaks or weekend vs. weekday patterns.

Controls: Pollutant selection, normalise to 0–1 scale (useful for comparing very different concentration levels), and facet type.

Calendar Plot

This chart displays a full calendar year where each day is coloured by pollution level - similar to a heat map on a calendar. Dark red days are more polluted; green days are cleaner. Seasonal patterns and pollution episodes stand out immediately.

Controls: Pollutant, year, statistic (mean, max, min, median), annotation style, colour palette, and optional WHO/AQI break points.

5.4 Wind & Rose Category

NOTE: All Wind & Rose plots require wind speed (ws) and wind direction (wd) columns in your data. These are included automatically when you fetch OpenAQ data that contains meteorological sensors. If your data lacks wind columns, these plots will display a warning.

Wind Rose

A wind rose shows the frequency of wind from each compass direction, with the length of each 'petal' indicating how often wind blows from that direction and the colour indicating the wind speed range. Use it to understand the dominant wind patterns at your station.

Pollution Rose

This combines a wind rose with pollution concentrations. The petals are coloured by pollutant level rather than wind speed. When longer petals are in a consistent direction, it suggests a pollution source in that direction.

Percentile Rose

Shows the 5th, 25th, 50th, 75th, and 95th percentile of concentration for each wind direction sector. This is more robust than the pollution rose because it is less sensitive to occasional extreme events.

5.5 Polar Category

Polar Plot

One of the most powerful tools in openair. The polar plot shows concentration as a function of both wind speed (radial distance from centre) and wind direction (compass angle). Hot spots in the chart indicate both the direction and the typical wind speed associated with high pollution, helping you pinpoint emission sources.

Polar Frequency

A polar frequency chart shows the distribution of observations across wind speed and direction categories, weighted by concentration or simply by count. Useful for understanding how frequently different wind regimes occur.

Polar Annulus

The polar annulus arranges time along the radial axis (hours, months, or seasons) and wind direction along the compass angle. The colour shows concentration. This reveals how pollution varies with both direction and time simultaneously.

5.6 Trend Category

Smooth Trend

Fits a smooth curve (with confidence interval bands) to time-averaged data. This is ideal for visualising long-term changes in air quality while filtering out day-to-day noise. The grey ribbon around the line represents the uncertainty in the trend estimate.

Theil-Sen Trend

A statistically robust trend test that is resistant to outliers. It reports the rate of change in concentration per year along with a p-value (probability that the trend occurred by chance). A p-value below 0.05 is generally considered a statistically significant trend.

Trend Level

A 2-D heat map showing concentration across two-time dimensions simultaneously - for example, hour of day on the vertical axis and month of year on the horizontal axis. Useful for identifying whether pollution peaks occur at the same time every day regardless of season, or whether the pattern changes.

5.7 Model Evaluation Category

Scatter Plot

A standard X vs. Y scatter plot. Use this to compare two pollutants or to compare measured vs. modelled data. The optional smooth fit line shows the overall relationship.

Model Statistics

Calculates numerical performance metrics used in air quality modelling: mean bias (MB), normalised mean bias (NMB), root mean square error (RMSE), correlation coefficient (r), and the fraction of values within a factor of two (FAC2). Select an observed column and a modelled column from your data.

5.8 Generating and Saving Plots

Generating a plot:

49. Select the plot type from the drop-down.
50. Adjust the controls in the Plot Controls panel.
51. Click the green Generate Plot button.
52. The chart appears in the right panel within a few seconds.

Saving a plot:

Three download buttons appear below the Plot Controls panel:

- PNG - a high-resolution image file suitable for reports and presentations
- PDF - a vector PDF suitable for publication-quality printing
- SVG - a scalable vector file suitable for web use and further editing in Inkscape or Illustrator

TIP: For maximum quality in printed reports, always use the PDF format. PDF plots are resolution-independent and will look sharp at any size.

Chapter 6: Export

The Export tab consolidates all download options in one place and adds additional metadata exports.

6.1 Dataset Exports

The left panel provides four buttons to download the main data table:

- CSV - comma-separated values, compatible with Excel and Google Sheets
- RDS - R binary format for direct use in R scripts
- Excel - native Excel .xlsx format
- ZIP - a single archive containing both CSV and RDS

6.2 Metadata Exports

The right side of the left panel provides three additional downloads:

- Locations - a CSV of all stations found in your last search
- Sensors - a CSV of the sensor details for the selected station

- Fetch Summary - a plain-text log of when you fetched the data, which station, which sensors, and how many rows were retrieved

6.3 Dataset Summary Card

The right panel shows a summary card of the currently loaded dataset, including the station name, number of rows, number of parameters, and the date range covered. This is a useful sanity check before exporting.

Chapter 7: Complete Worked Example

This chapter walks through a realistic analysis from start to finish. The example analyses PM2.5 air quality in Accra, Ghana, looking for patterns and trends.

Step 1 - Connect

53. Open API Data Retrieval & Analysis in your browser.
54. Paste your API key in the sidebar and click Connect.
55. Wait for the green 'Connected' badge.

Step 2 - Find a Station

56. Go to the Location Explorer tab.
57. Click the City / Name tab in the Search panel.
58. Type 'Accra' in the search box and click Search.
59. Several blue dots appear on the map over Accra, Ghana.
60. In the Found Locations list, click the card labelled 'Accra - US Embassy'.
61. The app highlights the station on the map and automatically takes you to the Fetch Data tab.

Step 3 - Fetch Data

62. In the Sensors panel, the available sensors are listed. Ensure pm25 is ticked.
63. Set Temporal Aggregation to Hourly averages.
64. Set the Date Range maybe to a full year (e.g. 2024-01-01 to 2024-12-31).
65. Set Max Records to 1000.
66. Click Fetch Data.
67. After a few seconds, a green message confirms the number of rows retrieved.

68. Scroll down to see the Data Preview table - it shows date, pm25, and any other available parameters.

Step 4 - Inspect the Data

69. Check the completeness bar for pm25. A green bar indicates good data coverage.
70. Review the table to confirm the date column is in a sensible format and values look reasonable.

TIP: PM2.5 values in an urban African city typically range from 15 to 150 $\mu\text{g}/\text{m}^3$. Values above 500 or below 0 may indicate sensor errors.

Step 5 - Prepare for Analysis

71. Go to the Analysis tab.
72. Ensure 'Use fetched OpenAQ data' is selected.
73. Click Prepare Data.

Step 6 - Summary Plot

74. From the Plot Type drop-down, select Summary Plot.
75. Set Averaging to 'day'.
76. Click Generate Plot.
77. A time series appears at the top showing daily pm25 values. The completeness bar at the bottom confirms data quality.

Step 7 - Time Variation

78. Select Time Variation from the Plot Type drop-down.
79. Ensure pm25 is selected in the pollutant list.
80. Click Generate Plot.
81. The four-panel chart reveals whether pm25 peaks in the morning rush hour, evenings, or certain days of the week.

Step 8 - Calendar Plot

82. Select Calendar Plot.
83. Set Pollutant to pm25, Year to 2024, Statistic to mean.
84. Tick 'WHO/AQI breaks (PM2.5/PM10)' to apply standard air quality categories.
85. Click Generate Plot.
86. The calendar shows which months were most polluted (e.g. harmattan season in November–February in West Africa).

Step 9 - Trend

87. Select Theil-Sen Trend.
88. Set Pollutant to pm25, Averaging to month.
89. Click Generate Plot.
90. The chart shows whether pm25 is increasing or decreasing over the year, with the slope expressed in $\mu\text{g}/\text{m}^3$ per year.

Step 10 - Export

91. Go to the Export tab.
92. Click the Excel button to save the full dataset.
93. Return to Analysis, generate your preferred chart, and click PDF to save a publication-quality plot.

Chapter 8: Reference Tables

8.1 All 15 Plot Types at a Glance

Plot Name	Category	Needs Wind?	Best Used For
Summary Plot	Time	No	First look; data quality overview
Time Plot	Time	No	Comparing pollutants over time
Time Variation	Time	No	Diurnal cycles, day-of-week patterns
Calendar Plot	Time	No	Seasonal patterns, pollution episodes
Wind Rose	Wind	Yes	Dominant wind directions and speeds
Pollution Rose	Wind	Yes	Likely source direction identification
Percentile Rose	Wind	Yes	Robust directional analysis (robust to outliers)
Polar Plot	Polar	Yes	Source identification; speed + direction + concentration

Plot Name	Category	Needs Wind?	Best Used For
Polar Frequency	Polar	Yes	How often each wind regime occurs
Polar Annulus	Polar	Yes	Direction vs. time-of-day or season
Smooth Trend	Trend	No	Long-term trends with uncertainty bands
Theil-Sen Trend	Trend	No	Statistically robust trend test
Trend Level	Trend	No	2-D heat map across two time dimensions
Scatter Plot	Model	No	Relationship between two variables
Model Statistics	Model	No	Quantitative model performance metrics

8.2 WHO Air Quality Guidelines (2021)

Pollutant	Annual Mean	24-hour Mean	Notes
PM2.5	5 µg/m ³	15 µg/m ³	Revised down from 10/25 in 2021
PM10	15 µg/m ³	45 µg/m ³	
NO2	10 µg/m ³	25 µg/m ³	Revised down from 40/200 in 2021
O3	-	100 µg/m ³	8-hour peak season guideline
SO2	-	40 µg/m ³	
CO	-	4 mg/m ³	24-hour mean

8.3 Common Error Messages and Solutions

Error	Likely Cause	Solution
401 Invalid API key	Key is wrong or expired	Re-copy from explore.openaq.org ; regenerate if needed
403 Access forbidden	Key does not have required permissions	Log in to openaq.org and check API key permissions
422 Validation error	Invalid parameter (e.g. radius > 25 km)	Reduce radius to 25 km or less
429 Rate limit	Too many requests in a short time	Wait 60 seconds and try again
No locations found	Spelling or area has no data	Try different spelling, coords search, or wider bounding box
No measurements returned	Sensor inactive in chosen period	Widen date range; try daily instead of raw aggregation
Wind rose blank	No ws / wd columns in data	Fetch a station that has meteorological sensors
Plot error: column missing	Pollutant not in prepared data	Re-check Prepare Data step; select the correct columns

Chapter 9: Tips and Best Practices

9.1 Getting the Best Data Quality

- Always check the completeness bars before starting analysis. Data with fewer than 50% valid values will produce unreliable charts.
- Prefer hourly or daily averages over raw measurements for analysis. Raw data includes more noise.
- If a station shows very few sensors, it may be a low-cost sensor rather than a reference monitor. Reference monitors tend to have better data quality but may update less frequently.
- Use the Max Records setting generously (5,000–10,000) to ensure you capture the full period of interest.

9.2 Choosing the Right Plot

- Start with the Summary Plot - always.
- Use Time Variation to answer 'when does pollution peak?'
- Use Calendar Plot to answer 'which season is worst?'
- Use Polar Plot to answer 'where does pollution come from?'
- Use Theil-Sen Trend to answer 'is air quality improving or worsening?'
- Use Scatter Plot to answer 'does PM2.5 rise when NO2 rises?'

9.3 Working with the Dark Theme

The default dark terminal theme uses black backgrounds and cyan/green text. When exporting plots in this theme, the chart backgrounds are white (openair draws on white by default), so exported plots will look professional regardless of the on-screen theme. The theme only affects the app interface, not the charts.

9.4 Citing Your Data

If you publish or share results, always cite:

94. OpenAQ as the data source: OpenAQ. (2024). Openair quality data platform. <https://openaq.org>
95. openair if you used the analysis tools: Carslaw, D.C. and Ropkins, K. (2012). openair - An R package for air quality data analysis. Environmental Modelling & Software, 27-28, 52-61.
96. The specific monitoring authority that collected the data (visible in the station details).

Chapter 10: Glossary

Term	Definition
API	Application Programming Interface - a standardised way for software to communicate over the internet
API Key	A unique secret code that identifies you to the OpenAQ server, similar to a password
Aggregation	The process of combining many individual readings into a single summary value (e.g. all readings in an hour averaged to one value)
BC	Black Carbon - soot particles produced by incomplete combustion, particularly from diesel engines
Bounding Box	A rectangular geographic area defined by its four corner coordinates (north, south, east, west boundaries)

Term	Definition
CO	Carbon Monoxide - a colourless gas produced by incomplete combustion
Confidence Interval (CI)	A range of values that likely contains the true value; a wider CI = more uncertainty
CSV	Comma-Separated Values - a plain text file format where columns are separated by commas, readable by Excel
Diurnal	Relating to changes that occur over the course of a single day (e.g. a diurnal cycle shows how values vary hour by hour)
FAC2	Fraction of values within a factor of two - a model evaluation metric; good models have FAC2 above 0.5
Latitude	North-south position on Earth's surface; 0° is the equator, +90° is the North Pole, -90° is the South Pole
Longitude	East-west position on Earth's surface; 0° is the Greenwich Meridian, ranges from -180° to +180°
NMB	Normalised Mean Bias - model bias as a percentage; 0% is perfect, positive means the model overestimates
NO	Nitric Oxide - a traffic emission precursor to NO2
NO2	Nitrogen Dioxide - a traffic and industrial emission associated with respiratory disease
NOx	Nitrogen Oxides - a collective term for NO and NO2
O3	Ozone - a secondary pollutant formed by sunlight reacting with traffic emissions; associated with smog
openair	An open-source R package for atmospheric data analysis developed at King's College London
PDF	Portable Document Format - a file format that preserves layout; ideal for printing and sharing
PM1	Particulate matter smaller than 1 micrometre in diameter

Term	Definition
PM2.5	Fine particulate matter smaller than 2.5 micrometres penetrates deep into the lungs and bloodstream
PM10	Coarse particulate matter smaller than 10 micrometres causes lung irritation
RMSE	Root Mean Square Error - a measure of typical error magnitude; lower is better
RDS	R Data Serialisation format - a binary file that stores R objects; opens directly in R with readRDS()
Sensor	An individual measurement device within a monitoring station; one station may have multiple sensors measuring different parameters
SO2	Sulphur Dioxide - emitted by volcanoes and burning fossil fuels; causes acid rain and respiratory disease
Station	A physical monitoring site, also called a Location in the OpenAQ database
UTC	Coordinated Universal Time - the global time standard used by the OpenAQ database; adjust to your local time zone as needed
WHO	World Health Organisation - sets international air quality guidelines
Wide Format	A data table where each column is a variable and each row is one observation in time - required by openair functions
ws	Wind speed - measured in metres per second (m/s)
wd	Wind direction - measured in degrees; 0° and 360° are north, 90° is east, 180° is south, 270° is west

API Data Retrieval & Analysis User Manual

Built with openair, R Shiny, leaflet, and the OpenAQ v3 API

Data: CC BY 4.0 - OpenAQ.org • openair: Carslaw & Ropkins (2012)